Insurance & Risk — Cheat Sheet

Financial Algebra · Unit 8: Insurance & Risk

Essential vocabulary

Premium
The amount paid regularly to keep an insurance policy active, whether or not you file a claim

Deductible
The amount you pay for covered costs before the insurer starts paying

Copay
A fixed amount paid for a covered service, such as a doctor visit

Coinsurance
Your percent share of covered costs after the deductible is met

Out-of-pocket maximum
The most you pay for covered care in a plan year; after that, the plan pays 100% of covered costs

Claim
A formal request to an insurer to pay for a covered loss

Policy limit
The maximum amount an insurer will pay for a covered loss

Expected value
The probability-weighted average of all possible outcomes

Risk pooling
Many people pay premiums so the losses of the few who have claims can be covered

Liability insurance
Coverage that pays for injuries or damage you cause to others

Collision coverage
Auto coverage that pays to repair your own car after an accident, minus the deductible

Comprehensive coverage
Auto coverage for non-collision damage such as theft, hail, or hitting a deer

Network (in-network)
Doctors and facilities that have agreed to a health plan's negotiated prices

Key formulas

Claim payout	min(loss − deductible, limit)
Health out-of-pocket	min(bills, ded) + coins × (bills − ded) + copays, ≤ OOP max
Total yearly cost	premium × 12 + out-of-pocket
Expected value	Σ p × outcome
Fair premium (insurer)	expected claim × (1 + load)
Deductible break-even	$\frac{\text{extra deductible}}{\text{yearly premium savings}}$

Coverage map

Your health	Health insurance
Others you injure	Liability (auto, renters/home)
Your car in a crash	Collision
Theft, hail, deer	Comprehensive
Your belongings (renting)	Renters insurance

Worked example

1. Silver plan: $2,000 deductible, 20%, $6,000 max; premium $210/mo
2. $9,000 in bills: 2,000 + 0.2 × 7,000 = $3,400
3. Premiums: $2,520
4. Total yearly cost: $5,920

Health savings account (HSA)
A tax-advantaged account paired with an eligible high-deductible health plan to save for medical costs

Self-insure
Choosing to cover a possible loss with your own savings instead of buying insurance

Risk tolerance (insurance)
How much financial uncertainty a person can handle; affects deductible and coverage choices

Common misconceptions

- Lowest premium = cheapest plan.
- Insurance is wasted if you never claim.
- After the deductible, insurance pays everything.
- Always choose the lowest deductible.

Questions to ask before deciding

- What is the worst-case yearly cost?
- Can my savings cover the deductible?
- Are my doctors in network?
- Are my liability limits high enough to protect my savings?
- Which small risks can I self-insure?

Decision tips

- Insure catastrophes; self-insure small losses.
- Keep the deductible in your emergency fund.
- Review coverage yearly.
- Get several quotes.

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